METRIC
MIL-DTL-71053A(AR)
29 May 2020
SUPERSEDING
MIL-P-71053
22 June 1994

DETAIL SPECIFICATION

PAX-2A EXPLOSIVE COMPOSITION

MIL-DTL-71053A, dated 29 May 2020, is hereby reactivated and may be used for either new or existing design acquisition.

This specification is approved for use by the US Army Armament Center and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements, examinations, and tests for manufacture and acceptance of PAX-2A, a high energy pressable explosive composition.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-45444 - HMX

Comments, suggestions, or questions on this document should be addressed to: Commander, U.S. Army Armaments Center, ATTN: FCDD-ACE-QSA, Picatinny Arsenal, New Jersey 07806-5000, or emailed to usarmy.picatinny.ardec.list.ardec-stdzn-branch@mail.mil. Since contact information can change, you may want to verify the currency of this information using ASSIST Online database at https://assist.dla.mil.

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DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-286	-	Propellants, Solid: Sampling, Examination and Testing
MIL-STD-650	-	Explosive: Sampling, Inspection and Testing
MIL-STD-1916	-	DOD Preferred Methods for Acceptance of Product

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

2.2.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

NAVAL SURFACE WARFARE CENTER

WS 1141	-	Bis (2,2-Dinitropropyl) Acetal-Bis (2,2-Dinitropropyl) Formal, Mixture Of, (BDNPA/F)

(Copies of these documents are available online at https://corp.dc3n.navy.mil.)

2.3 Non-Government standards and publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM E 11	-	Standard Specification for Woven Wire Test Sieve Cloth and Test Sieves (DoD adopted).
ASTM E 300	-	Standard Practice for Sampling Industrial Chemicals.
ASTM D 5675	-	Standard Classification for Fluoropolymer Micropowders

(Copies of these documents are available online at http://www.astm.org/).

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Required inspections.

3.1.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.1.2 Conformance. A sample shall be subject to conformance inspection in accordance with 4.3.

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3.2 Materials.

3.2.1 HMX. HMX shall be in accordance with MIL-DTL-45444, Grade B Class 5. In addition, the mean particle size distribution at the 50th percentile shall be not less than 2.40 µm and not greater than 7.10 µm.

3.2.2 BDNPA/F. The acetal/formal shall be in accordance with WS 1141.

3.2.3 CAB. The cellulose acetate butyrate shall be CAB-381-20 (see 6.9).

3.2.4 Fluoropolymer micropowder. For Type II, the fluoropolymer micropowder shall be PTFE0121 in accordance with ASTM D 5675.

3.3 Chemical and physical properties. PAX-2A shall conform to the chemical and physical requirements in Table I.

TABLE I. Chemical and physical requirements for PAX-2A.

Property	Type I	Type II
HMX[1], weight percent (wt. %)	86.6 ± 1.0	86.1 ± 1.0
BDNPA/F[2], wt. %	8.2 ± 1.0	8.2 ± 1.0
CAB[3], wt. %	5.2 ± 1.0	5.2 ± 1.0
Fluoropolymer micropowder[4], wt. %	------	$0.5\ ^{+0.2}_{-0.0}$
Total volatiles, % max.	0.10	0.10
Dry bulk density, g/cm^3, min.	0.85	0.85
Insoluble particles		
Retained on U.S. Alternate Sieve[5] No. 30, #	------	0
Retained on U.S. Alternate Sieve[5] No. 40, #	0	------
Retained on U.S. Alternate Sieve[5] No. 60, #	≤5	------
Granulation, % passing U.S. Alternate Sieve[5] No. 6, wt. %, min.	99	99

[1] HMX in accordance with 3.2.1.
[2] BDNPA/F in accordance with 3.2.2.
[3] CAB in accordance with 3.2.3.
[4] Fluoropolymer micropowder in accordance with 3.2.4.
[5] U.S. Alternate Sieves shall be in accordance with ASTM E 11.

3.4 Workmanship. PAX-2A shall be uniform in quality and free of visible impurities and foreign material such as dirt, rust, paint, wooden particles or any other foreign objects so that the safety and reliability of the explosive are not compromised.

3.5 Energetic material approval. PAX-2A materials or manufacturing processes used therein shall be approved for their intended use by the relevant Government Authority (see 6.6).

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4. VERIFICATION

TABLE II. Requirements/verification cross reference matrix.

Method of verification 1 – Analysis 2 – Demonstration 3 – Examination 4 – Test						Classes of verification A – First article B – Conformance		
Section 3 Requirement	Description	Verification Methods				Verification Class		Section 4 Verification
		1	2	3	4	A	B	
3.1.1	First article	X		X	X	X		4.2
3.1.2	Conformance	X		X	X		X	4.3
3.2.1	HMX	X		X	X	X	X	4.4.1 and 4.5.2
3.2.2	BDNPA/F			X		X	X	4.4.2
3.2.3	CAB			X		X	X	4.4.3
3.2.4	Fluoropolymer micropowder			X		X	X	4.4.4
3.3	Composition	X		X	X	X	X	4.5.1
3.3	Total volatiles	X		X		X	X	4.5.3
3.3	Dry bulk density	X		X		X	X	4.5.4
3.3	Insoluble particles	X				X	X	4.5.5
3.3	Granulation	X		X	X	X	X	4.5.6
3.4	Workmanship			X	X	X	X	4.6
3.5	Energetic material approval			X		X		4.7

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see paragraph 4.2).
b. Conformance inspection (see paragraph 4.3).

4.2 First article inspection. When specified, a sample of PAX-2A shall be subjected to first article verification in accordance with Table III. The testing shall apply to each of three (3) consecutive batches that have been produced by the contractor using the same production process, procedures, and equipment as will be used in regular production. All materials shall be obtained from the same sources as will be used in regular production.

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TABLE III. First article acceptance tests.

	Examination or test	Requirement paragraph	Inspection method reference
Critical	None defined		
Major			
101	HMX	3.2	4.4.1 and 4.5.2
102	BDNPA/F	3.2	4.4.2
103	CAB	3.2	4.4.3
104	Fluoropolymer micropowder	3.2	4.4.4
105	Composition	3.3	4.5.1
106	Total volatiles	3.3	4.5.3
107	Dry bulk density	3.3	4.5.4
108	Insoluble particles	3.3	4.5.5
109	Granulation	3.3	4.5.6
110	Workmanship	3.4	4.6
111	Energetic material approval[1]	3.5	4.7

[1] Test required following change to process control document (see 6.5).

4.2.1 Rejection. If any sample from any batch fails to comply with any of the applicable requirements the first article quantity shall be rejected.

4.3 Conformance inspection.

4.3.1 Inspection lot formation. A lot shall consist of one or more batches of PAX-2A of one Type designation only, produced by one manufacturer, meeting all requirements of this specification, and produced using one set of operating conditions not to exceed 31 consecutive calendar days, unless otherwise stated in the solicitation or contract. Each batch shall consist of that quantity of PAX-2A that has been subjected to the same unit chemical and physical mixing process intended to make the final product homogeneous. Lot formation shall be in accordance with MIL-STD-1916, Formation and identification of lots or batches. PAX-2A lot numbering shall be in accordance with MIL-STD-1168.

a. HMX of one Grade and Class only, from not greater than one interfix lot number from one manufacturer only (see 6.8).
b. BDNPA/F from one manufacturer in accordance with the same specification revision under one continuous set of operating conditions (see 6.8).
c. CAB from one lot and one manufacturer in accordance with the same specification revision under one continuous set of operating conditions only (see 6.8 and 6.9).
d. Fluoropolymer micropowder from one lot and one manufacturer in accordance with the same specification revision under one continuous set of operating conditions only (see 6.8).

4.3.2 Classification of characteristics.

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4.3.2.1 Sampling. The tests described in 4.3.2.2 shall be performed on samples representative of the batch which were taken in accordance with ASTM E300, for solids. Approximately 500 grams shall be taken for each batch. If any sample fails to meet any test requirement, the batch represented by the sample shall be rejected.

4.3.2.2 PAX-2A.

Category	Examination or Test	Verification Level	Requirement Paragraph	Inspection Method
Critical	None defined			
Major				
101	HMX	4.3.2	3.2	4.4.1 and 4.5.2
102	BDNPA/F	4.3.2	3.2	4.4.2
103	CAB	4.3.2	3.2	4.4.3
104	Fluoropolymer micropowder	4.3.2	3.2	4.4.4
105	Composition	4.3.2	3.3	4.5.1
106	Total volatiles	4.3.2	3.3	4.5.3
107	Dry bulk density	4.3.2	3.3	4.5.4
108	Insoluble particles	4.3.2	3.3	4.5.5
109	Granulation	4.3.2	3.3	4.5.6
110	Workmanship	4.3.2	3.4	4.6
Minor	None defined			

4.4 Materials.

4.4.1 HMX. Verify by examination of objective evidence that HMX is in accordance with 3.2.1 except that alpha HMX content may be determined using X-ray diffraction.

4.4.2 BDNPA/F. Verify by examination of objective evidence that BDNPA/F is in accordance with 3.2.2.

4.4.3 CAB. Verify by examination of objective evidence that CAB is in accordance with 3.2.3.

4.4.4 Fluoropolymer micropowder. Verify by examination of objective evidence that fluoropolymer micropowder is in accordance with 3.2.4.

4.5 Test methods and procedures.

4.5.1 Determination of composition.

4.5.1.1 Standard and sample preparation. Prepare standard and sample solutions of PAX-2A as follows.

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4.5.1.1.1 Standard preparation. Prepare 1 gram PAX-2A standard by weighing the appropriate masses of each individual ingredient (reagent-grade or better) and transferring to a 100-mL volumetric flask. Dissolve the ingredients in acetonitrile by ultrasonication with cooling for five minutes. Make up to volume with acetonitrile.

4.5.1.1.2 Sample preparation. Weigh approximately 1 gram PAX-2A sample and record to four (4) decimal places. Transfer to a 100-mL volumetric flask and dissolve in acetonitrile by ultrasonication with cooling for five minutes. Make up to volume with acetonitrile.

4.5.1.2 CAB composition. Using a Beckman FT1100 or equivalent Fourier transform infrared spectrophotometer, determine the absorbency of both the standard and the sample solutions prepared in 4.5.1.1 at approximately 1750 cm^{-1} (at the maximum wave number for alkyl carboxylates of cellulose acetate butyrate) using a 1.5 mm path length infrared cell. Use acetonitrile as the blank. Compute the weight % of CAB as follows:

$$(\mathrm{CAB})_s = \frac{A_s W_r (CAB)_r}{A_r W_s}$$

Where: A_s = net absorbency of sample solution
A_r = net absorbency of standard mixture solution
W_s = weight of sample, g
W_r = weight of standard mixture solution, g
$(CAB)_s$ = weight % of CAB in sample
$(CAB)_r$ = weight % of CAB in standard mixture solution

4.5.1.3 HMX and BDNPA/F composition. Use a High Performance Liquid Chromatograph (HPLC) with the following conditions:

a. HPLC: Varian Vista 5500, or equivalent.
b. Column: 22 cm RP8, 5 µm, 4.6 mm diameter
c. Detector: UV, 219 nm
d. Mobile phase: 35% water / 65% acetonitrile (v/v %)
e. Temperature: 35˚C
f. Flow rate: 1.5 mL/min
g. Injection volume: 10 µL (loop)

4.5.1.3.1 Procedure. Dilute 5-mL aliquots of both the standard and the sample PAX-2A solutions prepared in 4.5.1.1 with acetonitrile to 250 mL using a 5-mL pipette and 250-mL volumetric flasks. Determine the contents using the above conditions. Quantitation based on integrated area and the bracketing techniques are recommended. Compute the weight % of HMX and BDNPA/F as follows:

$$(HMX)_s = \frac{A_s W_r (HMX)_r}{A_r W_s}$$

$$(BDNPA/F)_s = \frac{A_s W_r (BDNPA/F)_r}{A_r W_s}$$

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Where: A_s = integrated area of HMX or BDNPA/F in sample solution
A_r = integrated area of HMX or BDNPA/F in standard mixture solution
W_s = weight of sample, g
W_r = weight of standard mixture, g
$(HMX)_s$ = wt. % of HMX in sample
$(BDNPA/F)_s$ = wt. % of BDNPA/F in sample
$(HMX)_r$ = wt. % of HMX in standard mixture
$(BDNPA/F)_r$ = wt. % of BDNPA/F in standard mixture

4.5.1.4 Fluoropolymer micropowder. Verify fluoropolymer micropowder content by subtracting the HMX, BDNPA/F, and CAB components from 100%.

4.5.2 HMX particle size. Particle size distribution shall be verified by use of either a Malvern or Microtrac particle size analyzer, or other equivalent laser particle size analyzer. Analytical procedures are as follows.

4.5.2.1 Malvern.

a. Riffle approximately 50 g of HMX.
b. Transfer approximately 0.5 g of the riffled material to a 50-mL beaker.
c. Add 10 mL of methanol to the beaker to serve as a wetting agent.
d. Add 15 mL of a surfactant solution (0.05% sodium metaphosphate or equivalent) to the beaker.
e. Ultrasonicate the sample for not less than five minutes.
f. Transfer sample to the Large Volume Sample Cell, containing 500-mL of distilled water, of the laser analyzer.
g. Ensure obscuration range for laser is between 0.3% and 3.0%.
h. Evaluate the sample with three runs each for particle size determination and record results.
i. Perform steps b through h two more times for three total samples.
i. Report the mean of the triplicate samples in step i for conformance.

4.5.2.2 Microtrac.

a. Place 50 g of HMX in a Gilson Rotating Riffler or equivalent and divide the sample into sixteen approximately equal portions.
b. Transfer one or more of these replicate portions into a beaker and blend with a spatula. Create 3 to 7 replicates from this beaker.
c. Take out approximately 0.5 g of one replicate and place into a 50-mL beaker.
d. Add one drop of Twitchell base (surfactant) to the beaker and manually mix with 0.5 mL heptane.
e. Add 20 mL heptane to the beaker. Sonicate for not less than five minutes.
f. Add 150±50 mL of filtered liquid to Microtrac Small Volume Circulator.
g. Quantitatively transfer approximately 1 mL aliquot of sample from the beaker to the circulator.

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h. Evaluate the sample with three runs each for particle size determination and record results.
i. Perform steps c through h two more times for three total samples.
j. Report the mean of the triplicate samples in step i for conformance.

4.5.3 Total volatiles. Determine the total residual volatiles in accordance with Method 101.6 Moisture and Volatiles (Vacuum Oven Method) of MIL-STD-650.

4.5.4 Dry bulk density. Determine the density of PAX-2A in accordance with Method 201.3 Specific Gravity (Mercury Displacement Method) of MIL-STD-650.

4.5.5 Insoluble particles. Determine number of insoluble particles in accordance with Method 106.1 Grit or Insoluble Particles (Solvent Extraction Method) of MIL-STD-650, except the solvent shall be either gamma-butyrolactone or n,n-dimethylformamide.

4.5.6 Granulation. Determine the granulation in accordance with Method 204.1 of MIL-STD-650 except 300 +15/-32 gyrations and sieves shall be in accordance with ASTM E11. Failure to meet the granulation shall result in batch rejection.

4.6 Workmanship. Take approximately 200 to 300 grams of PAX-2A and spread out on a clean white surface. The sample shall be examined with unaided eyes by personnel with normal vision or normal corrected vision. The sample shall be uniform in quality and free from lumps, grit, visible impurities, foreign matter or other defects that would render the material unsuitable for the intended use. Failure to meet the workmanship inspection and/or visual inspection of workmanship during the test methods of 4.5 shall result in lot rejection.

4.7 Energetic material approval. Verify by examination of objective evidence that the relevant Government Authority has approved all energetic materials for use (see 6.6).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point' packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

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6.1 Intended use. The material covered in this specification is intended to be used for application as an explosive where high energy and good thermal stability are needed in ammunition items.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Requirements for Acceptance Inspection Equipment (AIE) designs.
c. Whether a first article sample is required (see 3.1.1).
d. Packaging requirements, if applicable (see 5.1).
e. Propellant and Explosive Data Sheets in accordance with MIL-STD-1171.

6.3 Material safety data sheets. Contracting officers should identify those activities requiring copies of the completed material safety data sheets (SDS) prepared in accordance with FED-STD-313. The pertinent government mailing addresses for submission of data are listed in FED-STD-313, and 29 CFR 1910.1200 requires that the SDS for each hazardous chemical used in an operation must identify the activities requiring copies of the SDS.

6.4 Processing solvents. Ethyl acetate process solvent in accordance with TT-E-751 and methyl ethyl ketone process solvent in accordance with ASTM D 740, Type I, have both been found acceptable for use.

6.5 Process control document. A Process Document should be submitted to the Technical Agency (see 6.7) prior to the first article test. The document should contain a description of the process, all materials used, process conditions/procedures, and production equipment used to produce PAX-2A meeting the requirements of this specification. In addition, whenever there is a proposed change such as the source of materials, process conditions, procedures, etc., from those in the previous process control document, the Technical Agency (6.7) should be notified of the new process and process control change(s) and the reason for the change(s) with not less than 15 business days for Government approval prior to implementation.

6.6 Energetic material approval. Approval of energetic materials for Government contracts awarded to a manufacturer of PAX-2A requires demonstration that the energetic material meets the applicable requirements of AOP-7 and STANAG 4170. Point of contact for the Army Qualification Authority and list of qualified sources can be obtained from the Technical Agency (see 6.7).

6.7 Technical agency. Picatinny Arsenal is the technical agency that is responsible for the preparation and revision of the specification. All correspondence to the Technical Agency should be submitted through the contracting officer to: Commander, Armaments Center, ATTN: FCDD-ACE-QMG, Picatinny Arsenal, NJ 07806-5000.

6.8 Test and evaluation plan for approving new ingredient vendors. Table III provides guidance on those tests and test methods deemed necessary for qualifying a new ingredient vendor. The extent of testing recommended depends on the ingredient. Some programs may want to add system level testing for their application.

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6.9 CAB-381-20. The cellulose acetate butyrate found to be acceptable in PAX-2A formulation is CAB-381-20 from Eastman Chemical Company, 200 South Wilcox Dr., Kingsport, TN 37660 with the following typical properties:

TABLE IV. CAB-381-20 properties.

Property	Lower limit	Upper limit
Butyryl Content, weight (wt)%	35.00	39.00
Acetyl Content, wt%	12.00	15.00
Hydroxyl Content, wt%	1.40	2.20
Moisture, wt%	0	3.00
ASTM-A Viscosity, Ball Drop, s	17.000	28.000
Free Acidity by IC, PPM	0	300

6.10 Revision/ECP tracking. Revision A of this specification incorporated changes from ECP R04A2027 and made other changes for MIL-STD-961 compliance, and for clarity. ECP R04A2027 added Type II explosive, deleted the use of CAB-O-SIL and graphite, changed density requirement from pressed grain to dry bulk, removed physical form, and added the granulation requirement.

6.11 Subject term (key word) listing.

HMX
BDNPA/F
Main charge
Thermobaric

6.12 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian:
Army – AR

Preparing activity:
Army – AR
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